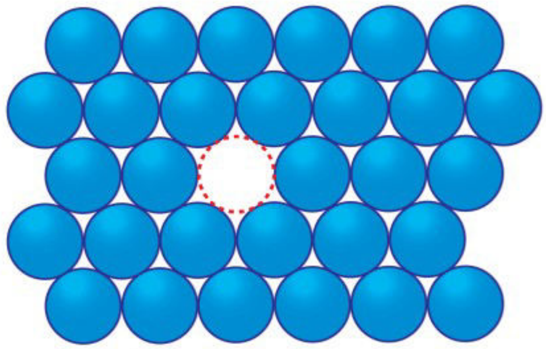
October 17, 2022

APS110 – Lecture 16

See hand written notes for drawings.

Imperfections

- Recall: impurity atoms can occupy interstitial sites or substitutional sites
- **Vacancies** → thermally generated



vacancy (missing atom)

- The atoms in a material all have different energy levels → ***on average*** all the atoms in the material have the temperature of kT
  - There is a distribution of energy levels
  - Only a **few** atoms have enough energy to form vacancies
- We can consider the number of atoms vs the energy level
  - When the temperature is lowered, there are more atoms in low energy states
    - At *absolute zero*, you would have only the lowest possible energy state occupied
  - What would happen at an *infinitely hot* temperature?
    - Would only the high energy states be occupied?
    - At infinitely high temp there needs to be increased entropy
      - If all the atoms would be in the highest possible energy state, this would actually be a more ordered state (entropy dictates how accurately you can guess the energy state of a randomly selected atom)
      - In this way, at infinitely high temperature there is still a distribution, trending towards all energy states **equally occupied**
  - Of course number vacancies are related to the number of atoms in high energy states
- Vacancies aren't that important to us with mechanical properties, but they are always present so we need to be aware of them

How can we use one-dimensional imperfections to **increase the strength** of a metal?

- LINEAR → '**DISLOCATIONS**'
- When you have plastic deformation, it means the dislocations have moved
- Plastic deformation also creates NEW dislocations

    - "Increase the dislocation density"
- Ex. slice of metal, 1mm x 1mm cross section
    - You can count the number of dislocation lines running through the slice
    - The number in this slice depends on the thermal history of a metal
    - If you carefully anneal the metal, to remove as many dislocations as possible, you can have ~1000 per mm^2
        - Anneal → heat treating to relieve stored plastic strain energy, reducing dislocation density
        - There are 'positive' and 'negative' dislocations, that can cancel each other out (dislocation annihilation)
    - For a heavily cold worked metal (very plastically performed) you can have ~10^7
    - There are a LOT of dislocations → by plastically deforming a metal we create a lot more (orders of magnitude)s
- When **dislocation density** is <u>high</u> → lots of dislocations, these dislocations run into each other and make it hard for the dislocations to move
    - <u>Inhibit</u> each others' movement → **increased strength**
    - Recall: the dislocations have strain fields around them

Two-dimensional imperfections

- <u>Note</u>: these imperfections can be used to increase strength, but they can change other properties as well – optical properties, electrical properties
- Grain boundary (grain = crystal)
    - As a metal is in the liquid state, eventually some of the atoms will start to slow down enough to form a crystal
    - Somewhere else in the liquid, another group of atoms will slow down enough to form a crystal, but they will be in a different orientation
    - Thus the crystal plane orientation tends to differ across grain boundaries (hard for disloc. to move across)
- The planes of crystals <u>across grain boundaries</u> do not generally line up → this means that the dislocation needs to *step up* to get across the grain boundary
- Additionally, the atomic spacing across the boundary is not at equilibrium spacing
    - Thus the **dislocation has difficulty navigating across grain boundary**
- These reasons all contribute to the grain boundary making the dislocations harder to move → strengthening
- To increase strength, we need to decrease grain size (more boundaries, more difficult for dislocations to move
    - This is called **grain size reduction**
    - This is not a trivial process, but as engineers we should consider the opposite situation → when heating a metal, the grains are going to increase in size and thus the metal will lose strength